PHY2506

Problem Set 1

Assigned: Friday Sept 25th; Due Friday Oct 16th, 2026

1. In deriving the analysis equation in the scalar example in Lecture 2 we assumed that the background and observations errors are uncorrelated, i.e., $\langle \epsilon_0 \epsilon_b \rangle = 0$. Show that the analysis error in this scalar example is uncorrelated with the innovation, i.e., $\langle \epsilon_a (\epsilon_0 - \epsilon_b) \rangle = 0$.

2. Consider the case in which the observations and background in our scalar example are biased, i.e.,

$$\langle \epsilon_b \rangle = b_b, \quad \langle \epsilon_0 \rangle = b_0.$$

Construct new variables that are unbiased as follows:

$$\tilde{x}_0 = x_0 - b_0$$
$$\tilde{x}_b = x_b - b_b.$$

(a) Form the analysis equation in terms of the errors using these new unbiased variables and show that $\langle \epsilon_a \rangle = \langle x_a - x^t \rangle = 0$.

(b) Find W that minimizes $\langle (\epsilon_a)^2 \rangle$.

3. Consider the following three distributions:

Uniform,

$$p(x) = \begin{cases} \frac{1}{2\sqrt{3}}, & \text{if } -\sqrt{3} \le x \le \sqrt{3} \\ 0 & \text{otherwise} \end{cases},$$

Triangular,

$$p(x) = \begin{cases} \frac{\sqrt{6}+x}{6}, & \text{if } -\sqrt{6} \le x \le 0 \\ \frac{\sqrt{6}-x}{6}, & \text{if } 0 < x \le \sqrt{6} \\ 0 & \text{otherwise} \end{cases},$$

Gaussian,

$$p(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{ -\frac{x^2}{2} \right\}.$$

(a) Plot these curves for x = [−4.0,4.0], with an interval of 0.1.

(b) What is the mean and variance for each distribution? Calculate this by hand, not numerically.

(c) Use the intrinsic random number generator in any analysis package that you are familiar with (e.g., MATLAB, Python, etc) to generate *n* random numbers with a uniform pdf. Plot the distribution of the random numbers for a number of different values of *n* (e.g., 10, 100, 1000, …). What is a value of *n* that will produce a good distribution of the pdf?

(d) Repeat Part (c) except for the normal distribution.

4. In this problem we generate a correlation matrix for a specific grid and examine its properties. Consider the interval $(-L_x, L_x]$, and divide it into J grid points. Similarly, divide the interval $(-L_y, L_y]$, with $L_y = L_x$, into J grid points. Consider the homogenous and isotropic, Gaussian correlation function in 1-D:

$$\rho(r = |x - y|) = \exp\left(-\frac{1}{2} \frac{(x-y)^2}{L_d^2} \right),$$

where r is the distance between any two pints in the domain and L_d is the decorrelation length. Points in the discrete domain are defined by

$$x_j = j\Delta x,$$

where $\Delta x = 2L_x / J$ for $j \in \{-J/2+1,\ J/2\}$, and corresponding values of y_j with $\Delta y = \Delta x$. The elements of the homogeneous, isotropic correlation matrix **B** are given by

$$B_{ij} = \rho(x_i, y_j)$$

(a) Construct **B** using (L_x, L_d, J) = (1, 0.2, 32). Make a contour plot of this the correlation array. (For example, in MATLAB and Python you can use "meshgrid" to generate this matrix.)

(b) For the parameters of Part (a), plot the correlation functions at the following two specific locations: $x_j \in \{0, L_x\}$

(c) Is the **B** obtained above an acceptable covariance matrix? Why or why not?

(d) From the figures constructed for Parts (a) and (b) we see that the correlations decrease very quickly toward zero. What if we actually set some of these small values to zero in order to save some storage space? Without actually changing the storage of the matrix, define a new matrix $\mathbf{B}_c$ by setting elements corresponding to $|r| > L_c$ to zero. Use the same parameters as in (a) and a cutoff value of $L_c = 3L_d$. Make a contour plot, and plot the correlation function at two locations as in Part (b). Is $\mathbf{B}_c$ an acceptable correlation matrix?

(e) Repeat Parts (a) to (c) for the Triangular correlation function,

$$T(x,y) = \begin{cases} 1 - |x-y| / L_c & \text{for } |x-y| < L_c \\ 0 & \text{otherwise} \end{cases}$$

(f) Construct a matrix **Q** as the Hadamard product of the matrices **B** and **T** by multiplying the matrices element by element:

$$\mathbf{Q} = \mathbf{B} \circ \mathbf{T} = [B_{ij} T_{ij}]$$

Make a contour plot of **Q**. Is **Q** an acceptable correlation matrix? Plot the correlation functions from **Q**, **B**, and **T** on the same graph, for $x = 0$.

5. Let us examine the case of two observations on a 1-D grid from Section 3.2 of the notes on Optimal Interpolation. As in the text, observation 1 is located at $x/L = -2.0$, the analysis is done at $x/L = 0$, and observation 2 varies over $x/L = [-4, 4]$. Assume $\alpha = (\sigma^r/\sigma^b)^2 = 0.25$ and

$$\rho(\Delta x) = \left(1 + \frac{|\Delta x|}{L}\right)\exp\left(-\frac{|\Delta x|}{L}\right).$$

Use Python script **pset1p5.py**. This script plots the weights w_1 and w_2 from Equations (3.20) and (3.21) as well as the normalized analysis error for Equation (3.22). Running **pset1p5.py** will reproduce Fig 3.2 of the notes (Fig 4.7 of Daley (1991)). In this problem, we consider how changes to the position of observation 1 and the value of α affect the results.

i) Consider the impact of changing the value of α in the code on the results. Try, for example, $\alpha = 0.1, 0.25, 0.5, 1.0$.

(a) What happens to the analysis error as α increases? Why?

(b) What happens to the phenomenon of observation screening?

ii) Consider how the results vary when observation 1 is moved. Plot the results when $\alpha = 0.25$ and observation 1 is located at $x/L = -3.0, -2.0, -1.0, 0.0$.

(a) When observation 1 is located at x_1, for what positions of observation 2 are the weights equal?

(b) What locations of observation 2 have no effect on the analysis?

(c) What is the best location for observation 1? For this location, over what length scale does observation 2 influence the analysis?

6. Filtering properties. In this problem, we are going to examine data assimilation as a form of filtering that is controlled by the error covariance matrices. Start by defining a grid over the intervals $(-L_x, L_x]$ and $(-L_y, L_y]$, with J grid points in each direction. (You can use the same domain parameters as in Problem 4a.) We will use the homogeneous and isotropic Gaussian correlation function in 1-D:

$$\rho(r = |x - y|) = \exp\left(-\frac{1}{2}\frac{(x-y)^2}{L_d^2}\right).$$

Here r is the distance between two points in the domain and L_d is the decorrelation length (with $L_d = 0.2$). Points in the discrete domain are defined by

$$x_j = j\Delta x$$

where $\Delta x = 2L_x/J$ for $j \in \{-J/2+1,\ J/2\}$, and the corresponding values of y_j with $\Delta y = \Delta x$. Form the homogeneous, isotropic correlation matrix **Q** using

$$\mathbf{Q}_{ij} = \rho(x_i, y_j).$$

(a) Find the eigenvalues and eigenvectors of **Q**. (You are free to use MATLAB, Mathematica, Python, or any statistical analysis package with which you are familiar.) Plot the eigenvalues as a function of index. Plot the eigenvectors corresponding to the 6 largest eigenvalues, as a function of *x*. Draw a zero-line and count the number of zero crossings for each eigenvector plotted.

(b) Now compute

$$\mathbf{A} = \mathbf{PH}^T(\mathbf{HPH}^T + \mathbf{R})^{-1} = \mathbf{P}(\mathbf{P}+\mathbf{R})^{-1}$$

where we have assumed **H** = **I**. Here $\mathbf{R} = (\sigma^r)^2\mathbf{I}$, $\mathbf{P} = \mathbf{DQD}$ and **D** is a diagonal matrix with each diagonal element equal to σ^b. Let $\sigma^r = 1$ and $\sigma^b = 2$. Compute the eigenvalues and eigenvectors of **A** making sure to sort by magnitude. Compare the eigenvalues of **A** to $\left(1+\alpha/\lambda_q\right)^{-1}$ where $\alpha = (\sigma^r/\sigma^b)^2$ and λ_q is an eigenvalue of **Q**. Plot the eigenvectors of **A**. Are they the same as for **Q**?

(c) Now let us create the innovations. To avoid interpolation, place an observation at every grid. To see the filtering aspects, define an innovation by

$$y = \cos(c\pi x)$$

where the *x*-grid values are from (-1,1]. Parameter *c* determines the wavenumber and hence the spatial scale of the observation increment. Create an innovation vector and compute the analysis increment using

$$\mathbf{d} = \mathbf{Ay}.$$

Plot **y**, **d** versus *x* for various values of *c*. Which waves are filtered the least? For which waves is the amplitude reduced by 80% or more?

(d) How are the filtering properties affected by a change in L_d?